ALI ASJAD SYED

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EDUCATION

Drexel University | Nick Howley College of Computing & Engineering — Philadelphia, PA
*B.S. in **Computer Engineering (Electrical and Computer Engineering)*** — *Expected Graduation: June 2028* | ***GPA: 3.78***

TECHNICAL SKILLS

Languages: C, C++, Python, SQL, MATLAB
Embedded & Hardware: Altium, STM32, CAN, SPI, Bluetooth, Oscilloscope, Logic Analyzer, Multimeter, Soldering
Robotics & Software: FreeRTOS, ROS, Linux, Git, KiCad, LTspice
AI/ML & Computer Vision: PyTorch, OpenCV

EXPERIENCE

Embedded Firmware & Low Voltage – Electrical Team — August 2026 – Present
Drexel Electric Racing — *Philadelphia, PA*
- Co-developed STM32 firmware (FreeRTOS) with CRC-checked SPI drivers reading 195 channels from daisy-chained ICs
- Debugged board-level SPI and CAN bus faults on the bench using an oscilloscope, logic analyzer, and digital multimeter
- Helped design the Altium schematic of a Teensy 4.1 DAQ board with 32 analog inputs over I2C and a CAN transceiver

Vision-Language-Action Machine Learning Models in Robotics — July 2026 – Present
Zhou Robotics Lab — *Philadelphia, PA*
- Applied OpenCV contour detection on 3 camera streams at 30 FPS to locate a marker pen and its lid for a 14-DoF robot
- Contributed to fine-tuning a VLA model on 29,804 frames from 3 camera views using JAX/PyTorch on the lab's GPUs
- Collected 61 teleoperated demonstrations through ROS on Linux and tested closed-loop control of all 14 joints over CAN

FP&A Data Analyst (Co-op) — March 2026 – September 2026
Barclays US Consumer Bank | Marlette Funding — *Wilmington, DE*
- Wrote SQL queries in Snowflake to consolidate 12 years of loan records for Power BI dashboards on 4 monthly KPIs
- Automated recurring Snowflake data pulls with Power Query and Python scripts to save 5 hours of reporting per week

RESEARCH & PUBLICATIONS

Second Author, ICLR 2027 Workshop Paper (Manuscript in Preparation) — July 2026 – Present
Improving Reinforcement Learning Post-Training for VLA Models Through Sequence-Level Policy Optimization — *Philadelphia, PA*
- Adapting Group Sequence Policy Optimization (GSPO) to VLA robot action sequences in the RLinf training framework
- Benchmarking GSPO against the GRPO baseline on robotic manipulation tasks for task success rate and training stability

Undergraduate Researcher, Brain-Computer Interfaces — October 2026 – Present
CONQUER Collaborative | Drexel University School of Biomedical Engineering & Science — *Philadelphia, PA*
- Supporting hardware setup of wearable fNIRS and EEG sensors for brain-computer interface studies with human subjects
- Filtering and classifying EEG and fNIRS signals with Python machine learning libraries to decode a user's commands

PROJECTS

PulseWatch – Heart Rate & Blood Oxygen Watch | *STM32, C, I2C, Bluetooth, Python* — June 2026 – August 2026
- Brought up an STM32 wearable with a MAX30102 optical sensor, OLED display, Li-Po charger, and Bluetooth module
- Updated the stock C firmware (STM32 HAL) to sample PPG signals at 100 Hz over I2C and stream the data via Bluetooth
- Implemented heart-rate and SpO2 algorithms in Python/MATLAB, within 2 bpm and 2 points of a store-bought monitor

Portable PCR Machine | *C++, PID Control, KiCad* — April 2026 – May 2026
- Modified the KiCad schematic of a USB-powered PCR machine and simulated its MOSFET heater driver in LTspice
- Refactored the Arduino firmware into object-oriented C++ PID and sensor classes with unit tests and Git version control
- Tuned the PID heater and PWM fan control to hold 95, 60, and 72 °C within 1 °C, checked with a reference thermocouple

Liquid Concentration Meter | *CircuitPython, I2C, KiCad, Python* — December 2025 – February 2026
- Assembled the device by hand-soldering the KiCad LED board and wiring a TSL2591 light sensor over the I2C bus
- Customized the CircuitPython firmware to stream light readings to a laptop over USB serial for live logging in Python
- Built Python calibration curves from 6 known dye concentrations, with readings repeatable to within 2% over 10 trials